

## Credit Risk Measurement and its Determinants: A Theoretical Approach

Bijoy Rakshit<sup>1</sup> Samaresh Bardhan<sup>2</sup>

### Abstract

Theoretically, this paper aims to analyze several credit risk models namely, probability of default (PD), loss given default (LGD), and exposure at default (EAD) and a few other relevant measures pertaining to credit risk management in Indian banking system. This paper also attempts to estimate PD and LGD of SME loans in one of the fast-growing banking markets, i.e. in India. Estimating these models essentially requires firm specific characteristics, macroeconomic variables and time effects. Rigorous estimates of credit risk models are crucial for initiating reserves for credit losses, calculating adequate risk capital and determining sound and vibrant policies for risky bank loans. While in India, most of the empirical literature by large dedicated into predicting the defaults and estimating losses on corporate bond markets, this paper will use a particular data set on firm's loan default in general and SMEs loan default in particular to estimate the credit risk. It is mandatory to acknowledge that although this paper highlights the relevant theories of credit risk model, its objectives, methodologies, selection of variables, data sources but the empirical findings of different credit risk models have not been reported in this paper due to intensive data requirements from various sources and annual reports of several credit rating agencies.

**Keywords:** probability of default (PD), loss-given-default (LGD), exposure at default (EAD) and SMEs loans,

---

<sup>1</sup> *Bijoy Rakshit, Doctoral Research Scholar, Department of Humanities and Social Sciences. Indian Institute of Technology, Ropar. Nangal Road, Punjab -14001. Email id: [bijoy.rakshit@iitrpr.ac.in](mailto:bijoy.rakshit@iitrpr.ac.in) Mobile No: 7508831411*

<sup>2</sup> *Dr. Samaresh Bardhan, Assistant Professor, Department of Humanities and Social Sciences. Indian Institute of Technology, Ropar. Nangal Road, Punjab -14001. Email id: [samaresh@iitrpr.ac.in](mailto:samaresh@iitrpr.ac.in) Mobile No:8288002217*

## 1. Introduction

Post the global financial crisis in 2008, measuring credit risk in financial and non-financial sector has become an indispensable area of research for both academicians and policy makers. The alarming financial fragility accompanied by deteriorated macroeconomic situation had led many banks (firms) into bankruptcy which in turn expedited massive business activity restrictions across the world. Hence, credit risk management has been a crucial issue in a successful banking and corporate practice and therefore, the issue has gained profound recognition from bank managers. Due to rapid proliferation of several credit risks (both systematic and unsystematic), more advanced and sophisticated risk management models have recently been introduced and been proved to be the most potent instruments to deal with credit risk management. Furthermore, it is hypothesized that the issue of credit risk is much susceptible during the phase of economic downturn and the effect of increasing credit risk may cause adverse effects when bank ex-ante overestimates the credit worthiness of their clients. Fiercer competition and unprecedented credit growth might induce banks to assign higher credit ratings to obligors, which can trigger problems in their portfolios when the macroeconomic situations deteriorate. But empirical estimation of credit risk models has drawn due attention only after the financial crisis. The necessity of early warning system (EWM) essentially came in to being with the motive to anticipate and identify the credit risk factors that contributes to financial fragility and destabilizes the entire economic system (Bussiere et al., 2006; Graciela et al., 1998).

Furthermore, efficient credit risk techniques considered to be the major instruments for commercial banks operating in fast growing economies. It is owing to surge in credit growth which significantly increases credit risk exposures in individual bank. Credit markets in Eastern European countries are an example of such developments. However, in recent times, the probability of default and other credit risk models are equally important in non-financial sector. As a result, a great deal of attention has been shifted to measuring firms' PDs in general and SMEs in particular and to link their credit exposures to banks. It is evident that models that include firm specific, macroeconomic and time varying factors are always best suited to explain the variations in credit risk models (volk, 2014). While linking estimated firms' PD to the

commercial banks, it has been found that in most cases both estimated PDs and credit ratings by banks exhibit completely different results for firm's credit worthiness. Thus, a separate demand has arisen to calculate the credit risk of banks as well as firms from both demand and supply perspective using financial as well as non-financial variables.

In the light of these considerations, this paper seeks to develop an early warning signal (EWS) model for predicting default in one of the quickly growing economies i.e. in India and explain several advanced and recent developed techniques of risk measurement namely - PD, LGD and EAD for both firms and banks along with firm-specific, macroeconomics and institutional variables. Second, this paper attempts to employ logistic model to directly estimate the probability of default. Additionally, this paper uses new Z-score model to highlight a high classification power on the estimated sample and a high predictive power in terms of its ability to detect bad firms in the sample. Third, this study explains the credit risk faced by the SMEs in India using CRISIL ratings; this particular rating agency has ratings on more than 25,400 large and mid-scale corporate and financial institutions in India<sup>3</sup>.

## **2. Definition and Methodology of Credit Risk Management**

Simplistically, credit risk can be defined as the potential that when a borrower fails to meet its obligations written explicitly in the terms and conditions of the organization from which they usually borrow. The sole aim of credit risk management is to maximize a bank's risk-adjusted rate of return by maintaining credit risk exposure within permissible limit. Banks are required to manage the credit risk that is inevitable to occur in the portfolio as well as the risk in individual credits or transactions. Therefore, the effective management of credit risk is very important and essential to the long run success of any banking organization.

Most of the default models are based on logit framework. The logistic model is a binary response model and is used where the probability of occurrence of an event is likely. The model can be used to give the probability that an event will occur given that the variables included in the

---

<sup>3</sup> CRISIL Annual Report, 2016

model are able to explain the event. This model can be borrowed to analyze the bank failure and its insolvency where ‘Y’ follows a Bernoulli distribution such that<sup>4</sup>

$$p\{Y_t | X\} = \begin{cases} 1 & \text{if bank in year } t \text{ is the year of default} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

One of the advantages of logit model is that it can anticipate the likelihood of a bank coming under the defaulted category based on explanatory variables. Following equation 2 (probability of default) and equation 3 (probability of not default), one can calculate the odds ratio. The odds ratio as written in equation 4 is the probability that a bank year in year t is a defaulted year divided by the probability that the year is not.

$$p\{Y = 1|X\} = \frac{e^{X\beta}}{1 + e^{X\beta}} \quad (2)$$

$$p\{Y = 0|X\} = 1 - \frac{e^{X\beta}}{1 + e^{X\beta}} = \frac{1}{1 + e^{X\beta}} \quad (3)$$

$$oddsratio = \frac{p\{Y = 1|X\}}{p\{Y = 0|X\}} = \frac{\frac{e^{X\beta}}{1+e^{X\beta}}}{\frac{1}{1+e^{X\beta}}} = e^{X\beta} \quad (4)$$

Taking natural logarithm of the odds ratio we get equation 5. While probabilities are restricted to values between 0 and 1 this transformation downs the logit model to values on R. It is worthwhile to mention that as the probability value tends to 0 the odds ratio also approaches towards zero implying that the event coded as default is very unlikely to occur and under such circumstances the logit model will tend to  $-\infty$ , conversely as the probability value tends to 1 both the odds ratio and the logistic transformation will tend to  $+\infty$ .

---

<sup>4</sup> Equations used for this model have been taken from the work of Mitchell, J Stock Forex Trad 2015, 4.2. The paper explains bank default prediction using a principal component analysis. The paper applies panel data on US banks for estimating loan default.

$$\ln(oddsratio) = \ln\left(\frac{\frac{e^{x\beta}}{1+e^{x\beta}}}{\frac{1}{1+e^{x\beta}}}\right) = \ln(e^{x\beta}) = x\beta \quad (5)$$

Shumway was of the view that most accounting variables tend to be highly correlated and as such the logistic accounting model often provides spurious results. In an attempt to make use of accounting variables and correct for multicollinearity it is imperative to use principal component analysis (PCA) methodology. PCA is essentially a variable reduction method widely used in face recognition software. Such methodology permits the researchers to reduce the number of variables in the model while retaining most of the information contained in those variables. The components or new variables that result from the exercise can then be used in the econometric analysis. Usually principal component can be defined as a linear combination of optimally weighted variables. A principal component for n variables is computed as follows:

$$C_i = \sum \beta_i X_i \quad (6)$$

$\beta_i$  is the coefficient for the variable  $X_i$  (given by solving an eigenequation).  $X_i$  is the first explanatory variable.

The principal components or the ‘new’ variables have certain specific characteristics. The first component will explain the majority of variations in the data and as such will be correlated with the explanatory variables. The second component will explain the variation that was left unexplained by the first component and this too will be correlated with the explanatory variables. Similarly, the third component will explain the variation that again was left unexplained by the first two components and this process will carry on until we have n components, with n being equal to the number of explanatory variables, explaining the total variation in the data. It is to be noted that while principal components are correlated with the explanatory variables, they are uncorrelated with each other.

### 3. An overview of PD, LGD and EAD models

Credit models related to PD, LGD and EAD are estimated by firms with information that reflects payment experience with the banks. The estimated parameters tend to explain individual bank experience and, in most cases, delineate different explanatory factors when compared to other banks' models. In order to provide a system-wide dimension to the risk parameters and to find out if there exist common risk factors across banks, three major sources of information are required:

1. Firms' credit statements that include loan information, information related to outstanding balance, rate of interest, actual payments, minimum required payment, and date of payment. Such information is collected to study the payment behavior of borrowers, identify recovery in subsequent periods after default and allow the identification of risk exposure at the time of default.
2. Public credit registry coverage and private credit bureau coverage that consists of individual credit records and contains information such as the total amount of loan and advances that the borrower had with other banks in the analyzed period, its performance and the time elapsed since the borrower first received a loan in the system.
3. Social housing institute information which collects payroll deductions from workers and describes borrowers' employment history and current income level.

#### 3.1. Probability of Default (PD) model

Usually categorical data techniques namely logistic regression is used in models while predicting default and this approach is proposed for the estimation of the probability of default (PD) model.

Let  $X_i'$  be a vector of  $p$  independent variables,  $y_i$  denotes the value of a dichotomous outcome variable, and  $i = 1, 2, 3, \dots, N$ . Furthermore, assume that the outcome variable has been coded as 0 or 1, denoting the occurrence or the non-occurrence of default respectively. To fit the logistic regression model requires the estimation of the vector  $\beta' = (\beta_0, \beta_1, \beta_2, \dots, \beta_p)$ .

Let the conditional probability that the outcome is present be denoted by  $p(Y = 1|X') = \pi(X')$ .

The logit of the multiple logistic regression model is given by the equation  $g(X') = \beta_0 - \beta_1 X_1 - \beta_2 X_2 - \dots - \beta_p X_p$ , in which case the logistic regression model is  $\pi(X') = \frac{e^{g(X')}}{1 + e^{g(X')}}.$

### **Explanatory Variables for PD model**

Following the study of Arindam Bandyopadhyay, (2006), these explanatory variables have been selected

- **Working Capital to Total Asset:** Working capital over total assets is a measure of the net liquid assets of the firm relative to total capitalization. Working capital is usually defined as the difference between current assets and current liabilities. So, this ratio of working capital to total assets can be used as a proxy variable for short the liquidity condition of firms under study.
- **Profit in cash to total ratio:** This is a measure of firm's cash flow. The standard formula for obtaining cash profit is to add the non-cash charges such as depreciation and amortization to the profit after tax, which is also known as net profit
- **Solvency Ratio:** Solvency ratio represents the long-run solvency of a firm. Higher is the solvency ratio better is the ability of the firm to meet key term obligations and lower is the chances of default. This is calculated by dividing the firms' total assets by its total borrowing plus current liabilities and provisions minus advance payment of tax.
- **Operating Profit to total assets:** operating profit over total asset is a measure of the true productivity of firm's assets. It measures the firm's earning capability. Higher the ratio, better the company is.
- **Sales to total asset:** The capital turnover ratio is a standard financial ratio. The firm's financial ratios like measure of liquidity, solvency, indebtedness, cash flow, profitability are the key elements of PD model. This ratio was used by Altman in his origin 1968 model implying the sales generating ability of the firms' assets. This ratio tells as to how

efficiently a firm is utilizing its assets. Smaller firms are less diversified and more financially constrained as compared to large firms and may have problems in raising funds during economic downturns. (Carling et al., 2007; Bernanke et al., 1996)

Apart from this financial variable, non-financial variables are taken from several existing literatures and it includes age of the firms' i.e. age of the firm since its incorporation, ownership groups, and other firm specific control variables(see Altman, 2000; Gangopadhyay et al., 2001; Bandyopadhyay and Das, 2005).

### **3.2. Modeling Loss Given Default (LGD)**

Developing an appropriate econometric model for Basel-complaint loss given default is very challenging. Starting from defining LGD, converting raw data for use in standard regression techniques, and defining variables largely requires a great deal of effort to estimate the model correctly. To estimate the model accurately a multifactor statistical model is required that can take in to account data at the loan level, the firm level, and the macroeconomic level.

Historically, LGD was estimated considering the historical averages, usually segregated by collateral type and seniority type. Siddiqi and Zhang (2004) offer an approach to calculate LGD, that regresses LGD on several independent variables and demonstrated which factors drive bank credit losses and, further the amount of losses to expect when loans to default. They incorporated macroeconomic, firm-specific, and loan specific information in to their model. Their findings revealed that several explanatory variables were both economically and statistically significant in modeling LGD. Overall descriptions of the dependent and independent variables are given below.

#### **Dependent variables:**

LGD is defined as the percentage of the charge-off (net of charge-off recovery) over outstanding balance at default for each defaulted loan. The definition of LGD requires huge care as there are many ways of defining it. Theoretically, loss derived from net charge-off should be the same as that derived from the remaining balance of total cash outflows and cash inflows after default. Specifically



Loss derived from net charge-off= charge-off-charge-off recovery

Loss derived from cash flows = total cash outflows - total cash inflows

As normally distributed variables are used to estimate OLS regression, one can transform LGD from its original underlying beta distribution to a normal distribution. This is done in two phrases.

First, we calculate the  $\alpha$  and  $\beta$  parameters from the underlying beta distribution as follows.

LGD = (Charge-off – charge-off recovery)/ outstanding balance at default

$\alpha$ : The beta distribution's shape parameter and can be derived from equations below

$\beta$ : The beta distribution's shape parameter and can be derived from equations below

Min: Minimum of all cases

Max: Maximum of all case

$\alpha$  and  $\beta$  are then derived from the following equations

$$\alpha = \frac{\mu}{Max} \cdot \left[ \frac{\mu (Max - \mu)}{Max \cdot \delta^2} - 1 \right] \quad (7)$$

$$\beta = \alpha \cdot \left[ \frac{Max}{\mu} - 1 \right] \quad (8)$$

Where  $\mu$  and  $\delta^2$  are population mean and variance respectively

In the second step, LGD is transformed from a beta to a normal distribution suitable for use in OLS regressions.

Using the definitions of  $\alpha$  and  $\beta$  as calculated above.

NLGD = Normal (LGD,  $\mu'$ ,  $\sigma'$ )

$$\mu' = \frac{\alpha}{\alpha+\beta} * Max \text{ and } \sigma' = \sqrt{\frac{\alpha*\beta}{(\alpha+\beta)^2(1+\alpha+\beta)}} * Max \quad (9)$$

## **Independent Variables**

The independent variables include loan-specific, firm-specific and macroeconomic information, as follows

Loan-specific information includes

1. Ratio of collateral value at default / outstanding at default
2. Ratio of collateral value at one year before default/outstanding at default
3. Outstanding balance at default /one year before default
4. Authorized balance at default
5. Risk rating at default
6. Risk rating at one year before default
7. Collateral type
8. Facility type

## **Firm-specific information**

- Leverage I: (total assets-net worth)/net worth.
- Leverage II: total debt/ capital at default
- Operating income/ sales at default
- Current liability/total liability at one year before default
- Firm size: revenue, total assets, net worth
- Firm age

## **Macroeconomic information**

- GDP quarterly growth rate one year before default
- Inflation

### 3.3. An alternative model of LGD

LGD is the credit loss when a borrower defaults and it is measured by the characteristics of the loan. This implies a common parameter in credit risk models which is commonly used in the determination of economic capital, anticipated loss or regulatory capital under Basel II for a bank institution. The credit risks are measured by the presence of collateral and when no collateral exists the cash flows that the borrower pays after default determine the LGD of the model. The model proposed to estimate LGD for the credit portfolio is to account for the cash flows that occur three months after default and compare them to the maximum outstanding balance of the loan after the moment of default.

The optimum outstanding balance can be used in order to consider the revolving nature of credit portfolios, and hence the possibility of balance increases due to line disposition in the period of default. Following this definition, LGD can be expressed

$$LGD = 1 - \frac{\sum_{i=t_{default}}^{t_3} \text{Borrower Payments}_i}{MAX\{Balance_{t_{default}}, Balance_{t_{default}+1}, Balance_{t_{default}+2}, Balance_{t_{default}+3}\}} \quad (10)$$

Where  $Bal_i$  is the outstanding balance of the credit portfolio at time “i” and  $t_{default}$  is the time of the default of the loan.

### 3.4. Empirical Analysis of EAD

One of the important issues associated with the analysis of credit risk is the measurement of EAD. EAD estimates the percentage of exposure the bank might lose if the borrower defaults. The calculation of EAD has become highly important in revolving instruments such as credit cards and therefore it is desirable to include an estimation of the value of the exposure that the borrower will have at the time of default in order to obtain an estimation of the expected loss. The significant difference between LGD and EAD is that unlike LGD, where risk is assumed to be calculated at a point after the default EAD is calculated very instantly as an account goes into default. However, default-time variables that can be used while modeling of LGD, cannot be used for estimating EAD. Only a few papers have looked at EAD for corporate loans, and even

fewer on retail loans. Araten and Jacobs (2001), Jimmez et al., (2009) and Jacobs (2008) in their literature focused mostly on the possible determinants of EAD and how EAD is affected by the economy and relationships between EAD and the behavioral of young firms.

Variables estimated in the literature are Exposure at Default Factor (EADF), the Credit Conversion Factor (CCF) and the Loan Equivalent Exposure (LEQ). All these variables are defined as ratios created using the outstanding balance at default over some indication of limit or balance at an observation time before default. However, it is possible to modify the variables slightly such that balance at default can be replaced by balance at some duration time  $t$ , and where the observation time is lagged  $I$  months, and the variables below are thus given in the most general form.

Suppose that, the outstanding balance of account  $i$  at time  $t$  is denoted by  $\beta_{it}$ , and limit of account  $i$  at time  $t$  is represented by  $L_{it}$ . To simplify the notation, the subscript  $i$  representing account  $i$  is dropped for the equations in this sub-section.

The  $EADF_t$  is the ratio of the balance at time  $t$  (at default), over the limit at observation time  $t-I$ , given in equation

$$EADF_t = \begin{cases} \frac{\beta_t}{L_{t-1}} & \text{for } L_{t-1} \neq 0 \end{cases} \quad (11)$$

The limit is set at the time of application to know the amount once the account is opened. Although, it is expected that  $EADF_t$  to range between 0 and 1, it is very much probable to observe balances greater than the assigned limits, perhaps because of accumulated interest or banks considering borrowers to go over their limits, providing values likely to be greater than 1.

The  $CCF_t$  is calculated as the ratio of the balance at time  $t$  (at default) over the balance at some observation time  $t-I$ , given in Equation

$$CCF_t = \begin{cases} \frac{\beta_t}{\beta_{t-I}} & \text{if } \beta_{t-I} \neq 0 \\ 0 & \text{if } \beta_{t-I} = 0 \end{cases} \quad (12)$$

$CCF_t$  is considered to be a better measure of predictions as it takes into account balance at default by measuring the outstanding balance of an account at some observation time before default

The  $LEQ_t$  measure of risk tries to make a more updated and sophisticated prediction for balance by not only taking into account balance at some observation time before default, but also the remaining amount of credit the debtor is able to draw upon. The equation for  $LEQ_t$  can be written as

$$LEQ_t = \begin{cases} \frac{\beta_t - \beta_{t-I}}{L_{t-I} - \beta_{t-I}} & \text{if } L_{t-I} \neq \beta_{t-I} \\ 0 & \text{if } L_{t-I} = \beta_{t-I} \end{cases} \quad (13)$$

The different values that the  $LEQ_t$  takes could arise due to a number of different situations and which would give different implications. It is predicted that the majority of accounts would have a positive  $LEQ_t$ , in such situation when, first, balance at default is greater than balance at observation, and balance at observation is below the credit limit at observation, which could be the most common progression into default and secondly when balance at observation is greater than balance at default, and balance at observation is already greater than the limit at observation, which would represent debtors who are actually recovering from a large balance.

Under certain circumstance it is possible to have negative  $LEQ_t$  especially when balance at observation is greater than limit at observation and balance at default turns out to be greater than balance at observation, which would represent debtors who are spiraling further into debt and

default. And when balance at observation is larger than balance at default, but both are below the limit at observation.

### **Explanatory and macroeconomic variables**

Common explanatory variables include firms' age and size, firms' identity such as time at address and time with bank, presence of landing and employment type. Certain behavioral variables of firms such as repayment amount, credit limit, number of outstanding loans, outstanding balance and number and value of cash withdrawals and credit transactions. Any behavioral variables used in this model should be lagged by certain time period. As far as macroeconomic variables are concerned average earnings index, index of production, all items retail price index can be used in this model.

### **4. Mixture Model for Outstanding Balance**

For predicting outstanding balance using a mixture model, we assume the random variable, balance of account  $i$  say at time  $t$  and the expected balance for account  $i$  at time  $t$  is given in equation (5)

$$E(\beta_{it}) = (p(\beta_{it} = L_{it}) \times E(L_{it} | \beta_{it} = L_{it})) + (p(\beta_{it} < L_{it}) \times E(\beta_{it} | \beta_{it} < L_{it})) \quad (14)$$

Since some accounts can have a value of  $B_{it}$  that is greater than the credit limit and we assume such accounts have an expected balance equal to the expected limit, we replace the first probability condition in Equation 5 by  $p(B_{it} \geq L_{it})$ . We therefore parameterize three models. First, a model of the probability that the outstanding balance of an account is larger than the credit limit; second, a model to predict the outstanding balance; and third, a model to predict the credit limit, where we allow parameters to predict balance and limit to differ.

Then we first estimate the probability that the outstanding balance at any duration time  $t$  is equal to or greater than the limit at  $t$ . This is done by defining the event 'overstretched',  $S_{it}$ , for account  $i$  and  $t$  which takes the value 1 if outstanding balance is greater than the limit at time  $t$ ; and 0 otherwise, given in Equation 5.

$$S_{it} = \begin{cases} 1 & \text{for } B_{it} \geq L_{it} \\ 0 & \text{otherwise} \end{cases} \quad (16)$$

Given this definition, it is possible for an account to experience the event more than once (at different times of the loan), so a discrete-time repeated events survival model, with clustered standard errors, is then used to estimate the model in equation 8

$$\log \left[ \frac{S_{it}}{1 - S_{it}} \right] = v + \beta_1 X_i + \beta_2 Y_{i,t-6} + \beta_3 Z_{t-6} \quad (16)$$

In equation 16,  $v$  represents the intercept term;  $X_i$  denotes account-dependent and time dependent covariates, i.e. application variables;  $Y_{i,t-6}$  are account- dependent, time- dependent covariates, lagged 6 months behavioral variables;  $Z_{t-6}$  are account - dependent, time-dependent covariates, lagged 6 months macroeconomic variables and  $\beta_1, \beta_2, \beta_3$  are unknown vectors of parameters to be estimated.

## 5. Structural Credit Risk Model

Merton (1974) develops a structural credit risk model where defaults are endogenously determined within the model. The model assumes that the default takes place if the value of assets falls short of the value of liabilities. However, one of the major shortcomings of the model can be attributed to the availability of market prices for the asset value. And such sort of information is always surprisingly scarce for small and medium sized enterprises. But few literatures show ways to overcome this problem with latent variable approach (Hamerle et al. (2003), Hamerle et al. (2004) and Rosch (2003)). They define the default even as a random variable  $Y_{it}$  which takes value 1 if firm  $i$  defaults in time  $t$  and 0 otherwise. The default can happen even when borrower's return on asset (ROA)  $R_{it}$ , falls short of some threshold  $c_{it}$ . the probability that a firm  $i$  will default in time  $t$ , given the survival until time  $t-1$  is described by the threshold model

$$\lambda_{it} = P(R_{it} < c_{it} | R_{it-1} \geq c_{it-1}) \quad (17)$$

Equation 17 can be explained with discrete time hazard rate model which gives the probability that firm  $i$  defaults in time  $t$  under the condition that it did not default before time  $t$ :

$$\lambda_{it} = P(T_i = t | T_i > t - 1) \quad (18)$$

As shown by Hamerle et al. (2003) it can always be assumed that the default event,  $Y_{it}$ , is observable. On the other hand, the observability of the return on firm's assets,  $R_{it}$ , depends on the available data. If  $R_{it}$  is observable then the model is linear. Otherwise a nonlinear model, such as logit or probit, is estimated, which treats the return on assets as a latent variable.

We estimate the probability that firm  $i$  defaults in year  $t$  given that it did not default in previous year  $P(T_i = t | T_i > t - 1)$  using different specifications of the model:

$$P(Y_{it} = 1 | X_{it}, Z_{it}) = F(\alpha + \beta X_{it} + \gamma Z_t) \quad (19)$$

Where  $Y_{it}$  is a binary variable which takes value 1 if firm  $i$  defaults in time  $t$  and 0 otherwise,  $\alpha$  is constant term,  $X_{it}$  is a vector of firm specific variables including also time invariant factors like sectoral dummies and  $Z_t$  is a vector of time varying explanatory variables, such as time dummies and macroeconomic factors.  $F(\cdot)$  is cumulative distribution function which is standard normal distribution function  $\Phi(\cdot)$  in the case of probit model and logistic distribution function in the case of logit model.

The estimated PDs can be used in comparison to credit ratings classification by banks. Banks have the ability to observe firm's riskiness in time  $t$  through monitoring and screening process and also can observe the state of the economy. The standard practice that banks follow while classifying borrowers in credit grades is credit overdue, which is available to banks regularly in time  $t$ . This means that in time  $t$  banks have a large set of information to take decisions about firm's credit worthiness. To make sure that same set of information is used as available to banks in time  $t$  we include all variables in the model at their values in time  $t$ , with few expectations. Finally, to estimate  $P(Y_{it} = 1 | X_{it}, Z_{it})$ , we apply random effects probit model. This estimator has already been used in other research and is the underlying model in Basel II risk assessment procedures. Hamerle et al. (2003) show that when only defaults are observable, an appropriate threshold model leads to random effects probit or logit model.



## 6. Conclusion

We have theoretically demonstrated several models for developing a reliable methodology to accurately examine the credit risk in Indian financial system especially for banks and SMEs loans. To make this study inclusive, we have considered firm specific characteristics and macroeconomic indicators in each model. However, apart from PD, LGD and EAD, two more advanced and sophisticated models namely structural credit risk model and mixture model of outstanding model have been used in this paper. All the models built are multifactor statistical model incorporating data at three levels: loan level, firm level and macroeconomic level in the final form. From policy perspective, establishing an appropriate credit risk environment is highly recommended so that the organizations can operate under a sound credit granting process and also can maintain an appropriate credit administration, measurement and monitoring process to ensure adequate control over credit risk. The role of supervisors and regulators should also be ensured.

## Reference

- Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The journal of finance*, 23(4), 589-609.
- Altman, E. I. (2000). Predicting financial distress of companies: revisiting the Z-score and ZETA models. *Stern School of Business, New York University*, 9-12.
- Araten, M., & Jacobs Jr, M. (2001). Loan equivalents for revolving credits and advised lines. *The RMA Journal*, 83(8), 34-39.
- Bandyopadhyay, A. (2006). Predicting probability of default of Indian corporate bonds: logistic and Z-score model approaches. *The Journal of Risk Finance*, 7(3), 255-272.
- Bandyopadhyay, A., & Das, S. K. (2005). The linkage between the firm's financing decisions and real market performance: A panel study of Indian corporate sector. *Journal of Economics and Business*, 57(4), 288-316.

- Bernanke, B. S., & Gertler, M. (1995). Inside the black box: the credit channel of monetary policy transmission. *Journal of Economic perspectives*, 9(4), 27-48.
- Bussiere, M., & Fratzscher, M. (2006). Towards a new early warning system of financial crises. *Journal of International Money and Finance*, 25(6), 953-973.
- Carling, K., Jacobson, T., Lindé, J., & Roszbach, K. (2007). Corporate credit risk modeling and the macroeconomy. *Journal of Banking & Finance*, 31(3), 845-868.
- Gangopadhyay, S., & Wihlborg, C. (2001). The Impact of Bankruptcy Rules on Risky Project Choice and Skill Formation under Credit Rationing.
- Hamerle, A., Liebig, T., & Rösch, D. (2003). Benchmarking asset correlations.
- Hamerle, A., Liebig, T., & Rösch, D. (2003). Credit risk factor modeling and the Basel II IRB approach.
- Hamerle, A., Liebig, T., & Scheule, H. (2004). *Forecasting credit portfolio risk* (No. 2004, 01). Discussion Paper, Series 2: Banking and Financial Supervision.
- Jacobs, K., & Li, X. (2008). Modeling the dynamics of credit spreads with stochastic volatility. *Management Science*, 54(6), 1176-1188.
- Jiménez, G., & Mencia, J. (2009). Modelling the distribution of credit losses with observable and latent factors. *Journal of Empirical Finance*, 16(2), 235-253.
- Kaminsky, G., Lizondo, S., & Reinhart, C. M. (1998). Leading indicators of currency crises. *Staff Papers*, 45(1), 1-48.
- Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *The Journal of finance*, 29(2), 449-470.
- Siddiqi, N. A., & Zhang, M. (2004). A general methodology for modeling loss given default. *RMA JOURNAL*, 86(8), 92-95.
- Volk, M. & Trefalt, P. (2014). Access to credit as a growth constraint. *Journal of Banking and Financial Economics*, 1, 29–39. doi: [10.7172/2353-6845.jbfe.2014.1.2](https://doi.org/10.7172/2353-6845.jbfe.2014.1.2)

